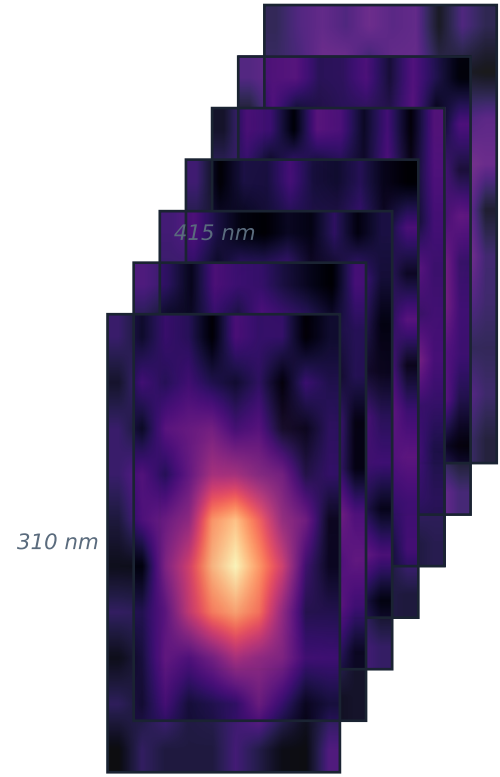


spectral-select · end-to-end pipeline

from raw 4-D ME-HSI cube to a ranked list of K (λ_{ex} , λ_{em}) bands

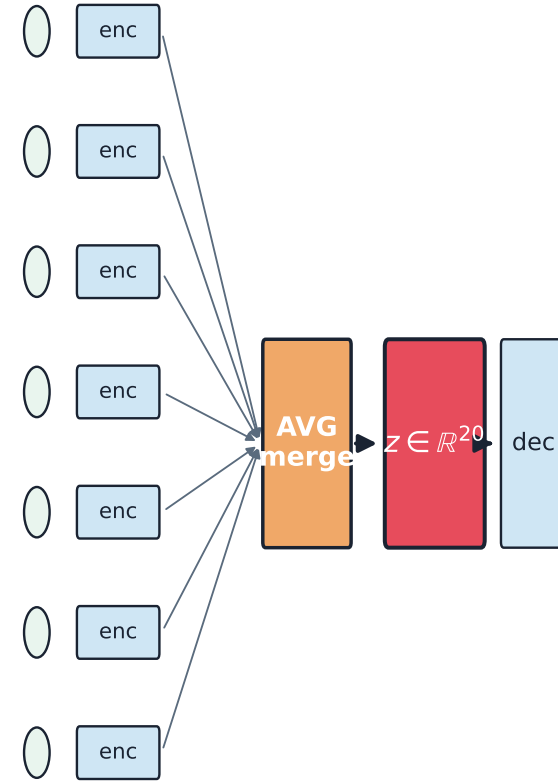
(A) ME-HSI cube
7 excitations × 31 emission bands



spatial × spectral × excitation

input cube

(B) 3-D CAE
parallel branches → shared latent



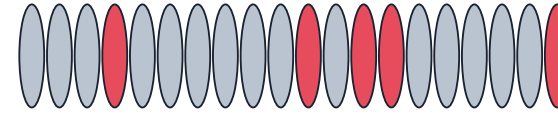
20-dim bottleneck forces shared structure

encode + bottleneck

(C) Latent perturbation

$$z'_d = z \pm \epsilon \sigma_d e_d$$

z (variance-ranked)

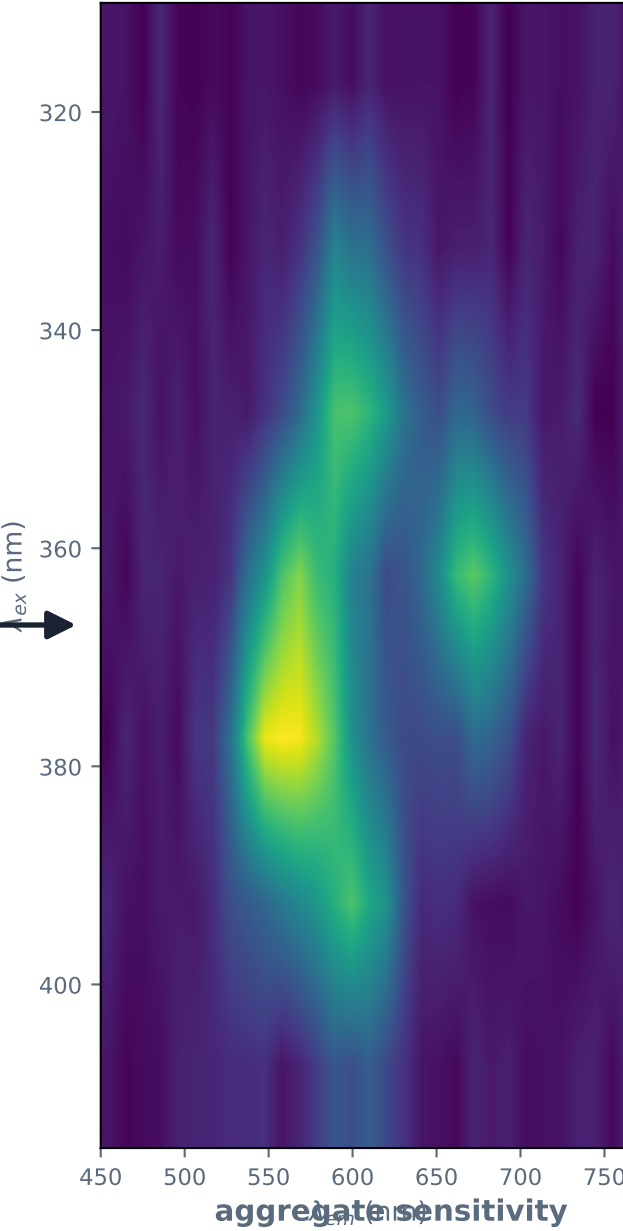


z' (perturbed top-k only)

→ measure $|\Delta \hat{X}(\lambda_{ex}, \lambda_{em})|$

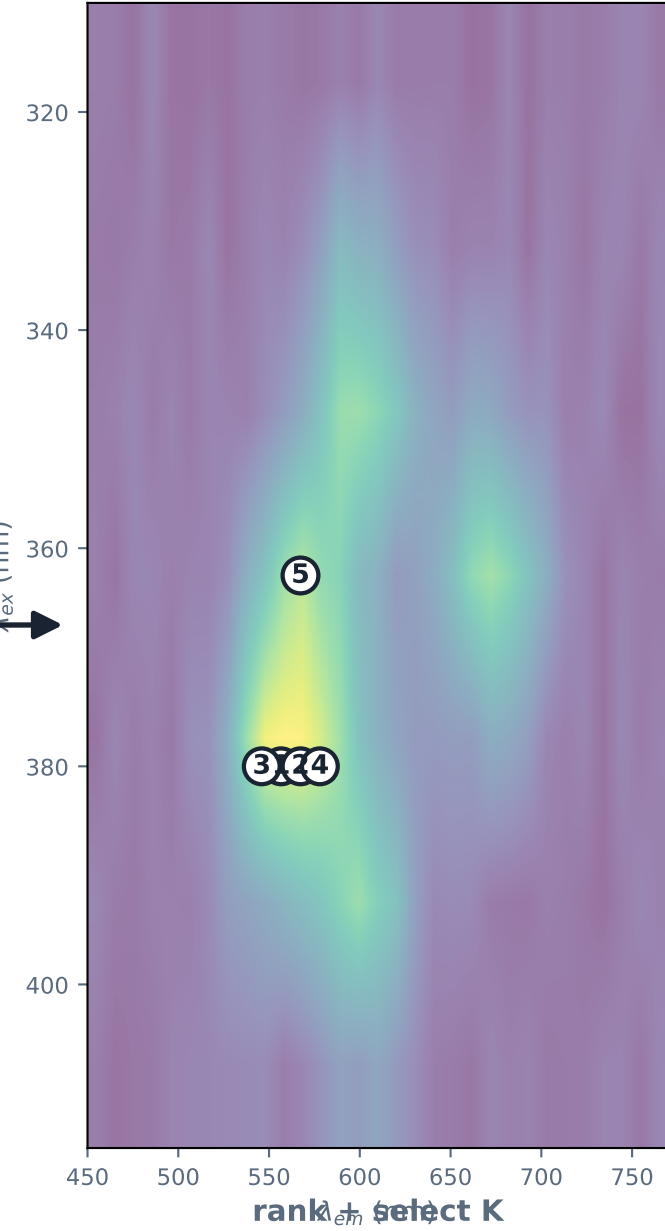
perturb latent

(D) Influence map
aggregated $|\Delta \hat{X}|$ over dims



aggregate sensitivity

(E) MMR → K bands
 $\lambda = 0.5$ (relevance ↔ diversity)



rank + select K

One forward pass per perturbation → influence matrix → ≤ 13 (λ_{ex} , λ_{em}) bands that match full-cube accuracy.